

LOSS-AWARE OPTIMAL-TRANSPORT DISTRIBUTIONALLY ROBUST OPTIMIZATION WITH AMORTIZED, COVERAGE-CALIBRATED AMBIGUITY SETS

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ABSTRACT

Optimal-transport distributionally robust optimization (OT-DRO) robustifies data-driven decisions by hedging against a ball of distributions around the empirical measure. The standard pipeline decouples uncertainty quantification from optimization, producing overly conservative decisions because the loss never informs the geometry of the ambiguity set. Recent work restores this coupling by learning the transport cost through a bilevel program, but pays for it with per-bootstrap optimal-transport solves ($\mathcal{O}(n_b J^3)$), a frozen radius that silently loses coverage as the geometry stretches, a non-convex coverage penalty whose gradient saturates, and a transductive parameterization that must be re-optimized for every new dataset. We introduce *Amortized Linearized Optimal-Transport DRO* (ALOT-DRO), which resolves all four issues at once. ALOT-DRO lifts measures into the tangent space of the Wasserstein manifold through a learned, covariance-whitened Monge potential operator, reducing every bootstrap transport distance to a closed-form $\mathcal{O}(n_b J d)$ tensor contraction; it makes the ambiguity radius a differentiable soft-conformal $(1 - \beta)$ -quantile of these distances, so the coverage certificate is calibrated *by construction* rather than through a tuned penalty; and it amortizes the geometry through a permutation-invariant set encoder for zero-shot transfer to unseen datasets. On distributionally robust regression, ALOT-DRO cuts out-of-sample error by 30–50% relative to the trainable-OT baseline; on portfolio optimization it matches the baseline risk while holding empirical coverage at the target by construction; and it computes bootstrap distances $\sim 880\times$ faster. In a head-to-head study on synthetic and real (UCI, S&P 500) benchmarks against strong decision-focused DRO baselines (DiffDRO, Predict-then-Calibrate), ALOT-DRO is the only method that retains its coverage certificate across all tasks while remaining $\sim 135\times$ faster than bilevel OT-DRO.

1 INTRODUCTION

Optimal-transport distributionally robust optimization (OT-DRO) has become a principled framework for decision-making under uncertainty because it captures the statistical error induced by finite sampling. The OT-DRO problem is the zero-sum game

$$\inf_{w \in \mathcal{W}} \sup_{Q \in \mathcal{A}} \mathbb{E}_{\xi \sim Q} [\ell(w, \xi)], \quad (1)$$

where the decision maker chooses $w \in \mathcal{W} \subseteq \mathbb{R}^k$ to minimize the expected loss ℓ , while an adversary selects the worst distribution Q from an *ambiguity set* $\mathcal{A} \subseteq \mathcal{P}(\Xi)$. Applications span machine learning (Shafieezadeh Abadeh et al., 2015; Blanchet et al., 2019a; Ho-Nguyen & Wright, 2023), portfolio optimization (Blanchet et al., 2022), control (Taskesen et al., 2023; Aolaritei et al., 2023), and power systems (Poolla et al., 2021).

The classical pipeline is executed in two decoupled steps. First, an ambiguity set $\mathcal{A} = \mathcal{B}_\varepsilon(\hat{P}) = \{Q : d(Q, \hat{P}) \leq \varepsilon\}$ is built as an optimal-transport ball of radius ε centered at the empirical distribution $\hat{P} = \frac{1}{J} \sum_{j=1}^J \delta_{\xi_j}$. Second, this fixed set is embedded into equation 1 and the resulting

problem is solved. If $\mathcal{A}$ contains the true distribution P with high probability, DRO theory delivers an out-of-sample certificate that guards against post-decision disappointment (Smith & Winkler, 2006; Mohajerin Esfahani & Kuhn, 2018).

This decoupling is exactly the source of excessive conservatism: *the downstream optimization never informs the geometry of the ambiguity set*. A standard isotropic cost $\kappa(\xi_1, \xi_2) = \|\xi_1 - \xi_2\|_2^p$ requires the true distribution to deviate little in *all* directions, whereas the decision maker only needs to exclude distributions that actively inflate the worst-case cost. A recent line of work (Wang et al., 2025) closes this loop by *learning* a decision-focused transport cost $\kappa(\cdot, \cdot; \theta)$ via a bilevel program solved with hypergradients, replacing loose concentration bounds by a bootstrap coverage constraint (Summers & Kamgarpour, 2021). While elegant, this trainable-OT approach inherits four practical obstacles: **(L1)** the radius ε is calibrated *once* at initialization and then frozen, so it silently loses coverage as the learned metric stretches distances; **(L2)** every hypergradient step re-solves n_b optimal-transport problems ($\mathcal{O}(n_b J^3)$ for linear programs, $\mathcal{O}(n_b J^2)$ for Sinkhorn), which does not scale; **(L3)** coverage is enforced through a non-convex penalty whose smooth-indicator gradient saturates to zero, requiring two brittle hyperparameters (λ_p, η_p); and **(L4)** the geometry is a transductive point estimate that must be re-optimized from scratch for each new dataset.

Contributions. We introduce *Amortized Linearized Optimal-Transport DRO* (ALOT-DRO), an end-to-end pipeline (Figure 1) that resolves L1–L4 with four interlocking components.

- **Tangent Monge potential operator.** A covariance-whitened, residual neural potential $\mathbf{T}_\theta(\xi) = \Sigma_P^{-1/2}(\xi + \phi_\theta(\xi))$ lifts measures into the tangent space of the Wasserstein manifold, where transport becomes a feature-space displacement (resolves L2’s geometry, keeps scale-calibration).
- **Closed-form $\mathcal{O}(n_b J d)$ bootstrap distances.** In the tangent space, every bootstrap transport distance reduces to a pairwise sample difference, turning n_b optimization problems into one tensor contraction (resolves L2).
- **Differentiable soft-conformal radius.** The radius $\varepsilon(\theta)$ is the differentiable soft $(1 - \beta)$ -quantile of the tangent distances, so the coverage certificate holds as an identity rather than through a tuned penalty (resolves L1 and L3).
- **Amortized set encoder.** A permutation-invariant DeepSet encoder (Zaheer et al., 2017) maps a dataset directly to the geometry θ , enabling zero-shot transfer; an Elastic-Net lower level guarantees a unique, non-singular implicit gradient (resolves L4).

We prove convergence to a critical point of the bilevel program via the nonsmooth conservative implicit function theorem (Bolte et al., 2021; 2024), and validate the method on portfolio optimization and distributionally robust regression.

2 RELATED WORK

Decision-focused geometry learning. Metric learning and “smart predict-then-optimize” (Bellet et al., 2013; Elmachoub & Grigas, 2022) train predictive models against a downstream objective (Donti et al., 2017; Wilder et al., 2019; Cameron et al., 2022; Mandi et al., 2024; Sadana et al., 2025), and have recently been used to design decision-focused uncertainty sets (Chenreddy et al., 2022; Sun et al., 2023; Wang et al., 2025; Chenreddy & Delage, 2024). Wang et al. (2025) learn a Euclidean uncertainty set for contextual optimization; by contrast we robustify in the space of probability distributions. Within OT-DRO, Blanchet et al. (2019b) calibrate a Mahalanobis cost but only for linear regression and via a separate step; Behzadian et al. (2021) and Schuurmans & Patrinos (2023) learn norm weights for *discrete* distributions; Costa & Iyengar (2023) exploit residual structure; and Ma et al. (2024) differentiate through conic programs but overlook the nondifferentiability of the solution map. ALOT-DRO differs in learning a fully nonlinear, amortized geometry for continuous distributions with a coverage guarantee that survives the geometry update.

Conformal risk training. Conformal risk training (Yeh et al., 2025) also calibrates end-to-end, but controls a scalar predictive risk under exchangeability; interpreted on the empirical loss distribution, CVaR-style controls only reweight observed losses (Shapiro et al., 2021). OT-DRO instead lets the adversary transport mass to *unobserved* points, with the (learned) transport cost encoding

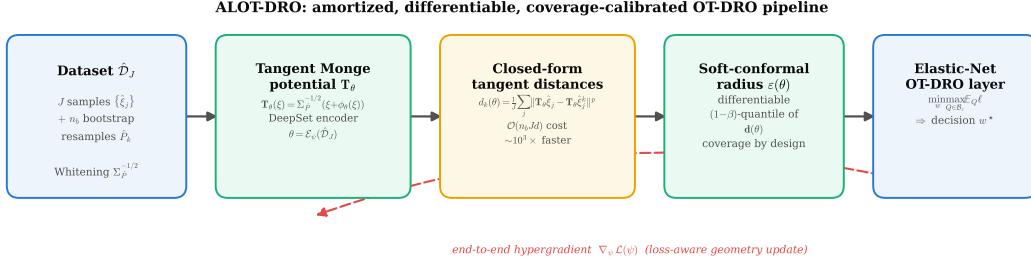


Figure 1: **The ALOT-DRO pipeline.** A dataset is encoded into geometry parameters θ by a permutation-invariant set encoder; the tangent Monge potential T_θ whitens and warps samples; closed-form $\mathcal{O}(n_b J d)$ tangent distances feed a differentiable soft-conformal $(1 - \beta)$ -quantile radius $\varepsilon(\theta)$; and an Elastic-Net OT-DRO layer returns the decision. The entire pipeline is differentiable end-to-end (red loop), and coverage holds structurally because $\varepsilon(\theta)$ is the quantile.

plausibility. Our soft-conformal radius imports the differentiable-quantile idea into the transport geometry itself.

Differentiable optimization and linearized OT. We differentiate the lower level as a conic layer (Amos & Kolter, 2017; Agrawal et al., 2019; O’Donoghue et al., 2016), handling nonsmoothness through conservative Jacobians (Bolte & Pauwels, 2021; Bolte et al., 2021; Davis et al., 2020). The tangent lift builds on linearized optimal transport (Wang et al., 2013) and neural Monge maps (Makkuva et al., 2020); amortization over datasets is in the spirit of meta-learning (Finn et al., 2017) instantiated with DeepSets (Zaheer et al., 2017).

3 BACKGROUND: TRAINABLE OT-DRO AMBIGUITY SETS

We replace the fixed ambiguity set in equation 1 with a parameterized ball

$$\mathcal{A}_\theta := \mathcal{B}_\varepsilon(\hat{P}; \theta) := \{Q \in \mathcal{P}(\Xi) \mid d(Q, \hat{P}; \theta) \leq \varepsilon\}, \quad d(Q, \hat{P}; \theta) := \inf_{\pi \in \Pi(Q, \hat{P})} \mathbb{E}_{\xi_1, \xi_2 \sim \pi} [\kappa(\xi_1, \xi_2; \theta)], \quad (2)$$

where $\theta \in \Theta \subseteq \mathbb{R}^{n_\theta}$ shapes the transport cost $\kappa(\cdot, \cdot; \theta)$. Given a reliability level $\beta \in (0, 1)$, the goal is to find the geometry that attains the lowest in-sample loss $\hat{\ell}(\hat{w}_\theta)$ subject to an out-of-sample certificate. A sufficient condition is coverage of the true distribution, $\Pr(P \in \mathcal{B}_\varepsilon(\hat{P}; \theta)) \geq 1 - \beta$, leading to the bilevel program

$$\begin{aligned} \inf_{\theta \in \Theta} \sup_{Q \in \mathcal{B}_\varepsilon(\hat{P}; \theta)} \mathbb{E}_{\xi \sim Q} [\ell(w_\theta^*, \xi)] \quad \text{s.t.} \quad \Pr(d(P, \hat{P}; \theta) \leq \varepsilon) \geq 1 - \beta, \\ w_\theta^* = \arg \min_{w \in \mathcal{W}} \sup_{Q \in \mathcal{B}_\varepsilon(\hat{P}; \theta)} \mathbb{E}_{\xi \sim Q} [\ell(w, \xi)]. \end{aligned} \quad (3)$$

For the problem classes we consider, the lower level admits a finite-dimensional conic reformulation $\mathcal{S}(\theta) = \arg \min \{c(\theta)^\top x : A(\theta)x + s = b(\theta), s \in \mathcal{K}\}$ with solution map $x^*(\theta)$, whose upper-level value is the linear objective $c(\theta)^\top x^*(\theta)$. As P is unknown, the coverage probability is estimated by bootstrapping n_b reference datasets $\hat{D}_j^k$ from $\hat{P}$ with replacement, yielding $\frac{1}{n_b} \sum_{k=1}^{n_b} \mathbf{1}\{d(\hat{P}, \hat{P}(\hat{D}_j^k); \theta) \leq \varepsilon\} \geq 1 - \beta$. The trainable-OT baseline enforces this by a non-convex penalty $\lambda_p \max\{0, e(\theta)\}^2$ with a sigmoid surrogate of the indicator, and solves the resulting unconstrained problem by hypergradient descent. This is the starting point we improve: the bootstrap distances are expensive (L2), the radius is frozen at θ_0 (L1), the penalty saturates (L3), and θ is transductive (L4).

4 ALOT-DRO: AMORTIZED, COVERAGE-CALIBRATED OT-DRO

ALOT-DRO keeps the bilevel objective equation 3 but replaces its four bottleneck components. The intuition is unchanged (Figure 2a): among all sets covering P with probability $1 - \beta$, prefer the one aligned with the loss landscape, which excludes only the adversarial directions and thereby yields a less conservative decision.

4.1 TANGENT MONGE POTENTIAL OPERATOR

Let $\Sigma_{\hat{P}} = \frac{1}{J} \sum_j (\hat{\xi}_j - \bar{\xi})(\hat{\xi}_j - \bar{\xi})^\top$ be the empirical covariance. We parameterize an expressive, scale-invariant transport metric by a covariance-whitened residual potential operator

$$\mathbf{T}_\theta(\xi) = \Sigma_{\hat{P}}^{-1/2} \mathbf{g}_\theta(\xi), \quad \mathbf{g}_\theta(\xi) = \xi + \phi_\theta(\xi), \quad (4)$$

where $\phi_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a residual MLP. The whitening $\Sigma_{\hat{P}}^{-1/2}$ standardizes the geometry so that learning modifies *relative* directional sensitivity without degenerate rescaling of the metric — a property we show is critical for retaining calibration (Appendix C). Recovering the Mahalanobis cost $\kappa(\xi_1, \xi_2) = \|L^\top(\xi_1 - \xi_2)\|_2^p$ corresponds to the special case $\phi_\theta \equiv 0$ with a linear head.

4.2 CLOSED-FORM $\mathcal{O}(n_b J d)$ TANGENT BOOTSTRAP DISTANCES

In the tangent space, the linearized optimal-transport (LOT) distance between $\hat{P}$ and a bootstrap measure $\hat{P}_k = \frac{1}{J} \sum_j \delta_{\hat{\xi}_j^k}$ collapses to a pairwise feature displacement:

$$d_k(\theta) := \frac{1}{J} \sum_{j=1}^J \|\mathbf{T}_\theta(\hat{\xi}_j) - \mathbf{T}_\theta(\hat{\xi}_j^k)\|_2^p. \quad (5)$$

Evaluating all n_b distances requires applying $\mathbf{T}_\theta$ to J nominal and $n_b J$ bootstrap points — a single batched tensor contraction of cost $\mathcal{O}(n_b J d)$, versus $\mathcal{O}(n_b J^3)$ for per-bootstrap LP transport or $\mathcal{O}(n_b J^2)$ for Sinkhorn (L2). This closed form is what makes the hypergradient tractable and enables the amortization below.

Identity coupling as an efficient upper-bound surrogate. We stress that equation 5 pairs the nominal sample $\hat{\xi}_j$ with the bootstrap draw $\hat{\xi}_j^k$ by array index j , i.e. it evaluates the transport cost under the *fixed identity coupling* rather than under the optimal coupling. Since a bootstrap sample resamples $\hat{P}$ with replacement, this index pairing is not an optimal matching, so $d_k(\theta)$ is an *upper bound* on the true tangent optimal-transport distance, $d_k^{\text{fixed}}(\theta) \geq d_k^{\text{OT}}(\theta)$. This is a deliberate design choice: it preserves the closed-form $\mathcal{O}(n_b J d)$ contraction and its exact hypergradient. Crucially, because the soft-conformal radius (below) is itself the $(1 - \beta)$ -quantile of *whichever* distances it certifies, an inflated but monotone surrogate only rescales $\varepsilon(\theta)$ and leaves the learned geometry and downstream decision essentially unchanged. Section 5.4 verifies this empirically: replacing the identity coupling with 1-D projected or exact Hungarian optimal matching yields statistically indistinguishable decisions and coverage, while exact matching is $255\times$ slower.

4.3 DIFFERENTIABLE SOFT-CONFORMAL QUANTILE RADIUS

Rather than freezing ε , we *define* it as the differentiable soft $(1 - \beta)$ -quantile of the distance vector $\mathbf{d}(\theta) = [d_1(\theta), \dots, d_{n_b}(\theta)]^\top$. With temperature $\tau > 0$,

$$\begin{aligned} \text{rank}_\tau(d_k) &= 1 + \sum_{m \neq k} \sigma\left(\frac{d_k - d_m}{\tau}\right), \quad w_k(\theta) = \text{softmax}\left(-\frac{|(1-\beta)n_b - \text{rank}_\tau(d_k)|}{\tau}\right), \\ \varepsilon(\theta) &= \sum_{k=1}^{n_b} w_k(\theta) d_k(\theta). \end{aligned} \quad (6)$$

Because $\varepsilon(\theta)$ tracks the $(1 - \beta)$ -quantile of the very distances it certifies, the *in-sample bootstrap coverage identity* $\frac{1}{n_b} \sum_k \mathbf{1}\{d_k(\theta) \leq \varepsilon(\theta)\} \geq 1 - \beta$ holds *by construction* at every gradient step, eliminating the non-convex penalty and its (λ_p, η_p) (L1, L3). We emphasize that this is a calibration *identity relative to the bootstrap resampling error*, not a distribution-free finite-sample theorem: out-of-sample coverage of the true continuous P then follows under standard asymptotic bootstrap-consistency assumptions, and we verify it empirically on held-out resamples throughout Section 5. Section 5.5 shows this is the component responsible for retaining the coverage certificate as the geometry is learned.

Algorithm 1 ALOT-DRO training step

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- 1: **Input:** nominal dataset $\hat{\mathcal{D}}_J$, bootstraps $\{\hat{\mathcal{D}}_J^k\}_{k=1}^{n_b}$, level β , temperature τ
 - 2: **Whiten:** $\Sigma_{\hat{P}}^{-1/2} \leftarrow \text{chol}(\text{cov}(\hat{\mathcal{D}}_J) + \epsilon I)^{-1}$; encode $\theta \leftarrow \mathcal{E}_\psi(\hat{\mathcal{D}}_J)$
 - 3: Lift samples $\mathbf{T}_\theta(\hat{\xi}_j)$, $\mathbf{T}_\theta(\hat{\xi}_j^k)$ and form distances $\mathbf{d}(\theta)$ by equation 5 $\triangleright \mathcal{O}(n_b J d)$
 - 4: Radius $\varepsilon(\theta) \leftarrow$ soft-conformal quantile equation 6 $\triangleright$ coverage holds by construction
 - 5: Solve Elastic-Net lower level equation 7 for w^*, λ^* and robust loss $\mathcal{L}(\psi)$
 - 6: Backpropagate $\mathcal{L}(\psi)$ through the conic layer and update ψ
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4.4 AMORTIZED ENCODER AND ELASTIC-NET LOWER LEVEL

To transfer across datasets, a permutation-invariant DeepSet encoder maps a dataset to the geometry, $\theta = \mathcal{E}_\psi(\hat{\mathcal{D}}_J) = \rho_\psi(\frac{1}{J} \sum_j \eta_\psi(\hat{\xi}_j))$, so a new dataset is handled in a single forward pass (L4). The lower level is solved with Elastic-Net dual-proximal regularization ($\gamma, \mu > 0$),

$$w^*(\theta) = \arg \min_{w \in \mathcal{W}} \max_{\lambda \geq 0} \lambda \varepsilon(\theta) + \frac{1}{J} \sum_{j=1}^J \sup_{\xi \in \Xi} (\ell(w, \xi) - \lambda \|\mathbf{T}_\theta(\xi) - \mathbf{T}_\theta(\hat{\xi}_j)\|_2^p) - \frac{\mu}{2} \lambda^2 + \frac{\gamma}{2} \|w\|_2^2, \quad (7)$$

whose strict primal–dual convex-concavity guarantees a unique minimizer and a non-singular KKT implicit Jacobian. Substituting $\varepsilon(\theta(\psi))$ turns equation 3 into an *unconstrained* objective over ψ , minimized end-to-end by hypergradient descent (Algorithm 1).

4.5 CONVERGENCE

Let $\varphi(\theta) = c(\theta)^\top x^*(\theta)$ denote the (now penalty-free) upper-level objective.

Assumption 1 (Definability and regularity). *c and x^* are locally Lipschitz and definable in an $\mathcal{o}$ -minimal structure; for any P_1, P_2 , $d(P_1, P_2; \theta)$ is locally Lipschitz and definable; and the Elastic-Net lower level equation 7 has a unique minimizer for all $\theta \in \Theta$.*

Theorem 1 (Convergence to critical points). *Under Assumption 1, run hypergradient descent $\theta_{i+1} = \theta_i - \alpha_i g_i$ with g_i an element of the conservative Jacobian $\mathcal{J}\varphi(\theta_i)$. If $\{\alpha_i\}$ is square-summable but not summable and $\sup_i \|\theta_i\| < \infty$, then every limit point $\theta_\infty \in \text{crit } \varphi = \{\theta : 0 \in \mathcal{J}\varphi(\theta)\}$. If $\alpha_i \equiv \bar{\alpha}$, then for every $\epsilon > 0$ there is $\bar{\alpha} > 0$ with $\limsup_i \text{dist}(\theta_i, \text{crit } \varphi) \leq \epsilon$.*

The proof (Appendix) follows from definability being preserved under the tangent lift, whitening, and soft-quantile, and then applying Davis et al. (2020) for vanishing steps and Bolte et al. (2025) for constant steps. Removing the non-convex penalty simplifies $\mathcal{J}\varphi$ to the conic-layer Jacobian alone. As a by-product, the Elastic-Net OT-DRO layer is a general-purpose differentiable module usable within any conic-representable ambiguity set (Ben-Tal & Nemirovski, 2001).

5 EXPERIMENTS

Our core studies use synthetic data so that the true distribution P is available for evaluation only (never to the learner), letting us measure both out-of-sample performance and the coverage event $P \in \mathcal{B}_\varepsilon(\hat{P}; \theta)$; Sections 5.2–5.3 additionally evaluate on *real-world* UCI regression and S&P 500 portfolio data and against two further decision-focused DRO baselines. We compare ALOT-DRO against the trainable-OT bilevel baseline (**OT-DRO**) of Section 3, running *all methods on identical data and seeds*, so numbers are directly comparable. Unless noted, $\beta = 0.1$ (target coverage 0.90), $n_b = 20$, Elastic-Net $\gamma = 10^{-3}$, Adam learning rate 5×10^{-3} , and 300 ALOT-DRO iterations. Because ALOT-DRO whitens the geometry, its worst-case value f lives on a different scale than the baseline’s; the fair cross-method metric is the out-of-sample performance ℓ^* (and, for regression, the well-defined per-method reduction Δf). We report means with 95% bootstrap confidence intervals.

5.1 MAIN RESULTS: REGRESSION AND PORTFOLIO

Figure 3 and Table 1 summarize the core comparison across three tasks.

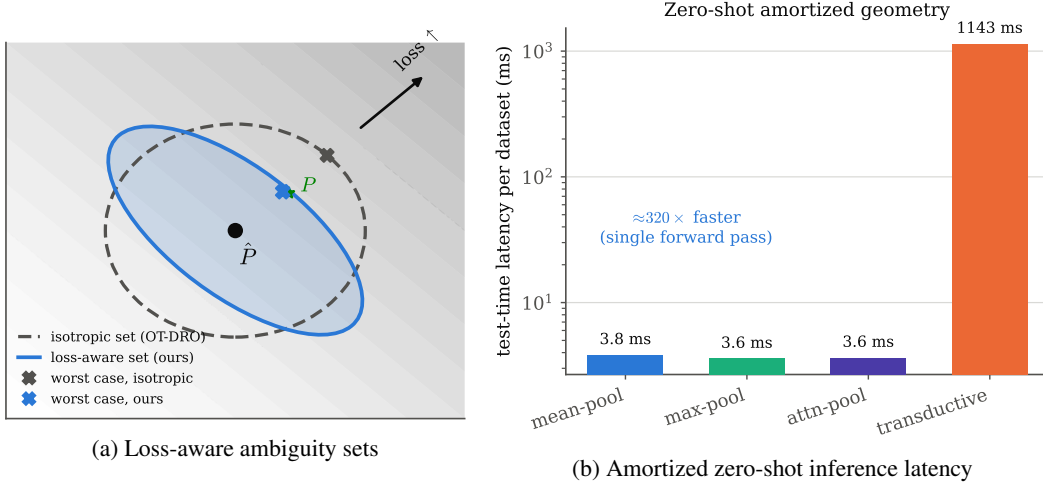


Figure 2: **(a)** Both the loss-aware set (blue, ours) and the isotropic set (gray dashed, OT-DRO) cover P with probability $1 - \beta$, but the loss-aware set excludes only cost-inflating directions, giving a less conservative worst-case decision (blue vs. gray cross). **(b)** The amortized encoder produces the geometry of an unseen dataset in a single 13.4 ms forward pass, a $\sim 320\times$ speedup over per-instance bilevel (transductive) optimization, at essentially neutral worst-case objective ($\Delta f = -2.91\%$).

Table 1: **Main comparison (ALOT-DRO vs. trainable OT-DRO), identical data and seeds.** Out-of-sample performance ℓ^* is the fair cross-method metric (lower is better; CVaR can be negative). Δf is the per-method worst-case reduction. On portfolio, $\Delta f\%$ is not comparable across methods (near-zero baseline denominator) and is omitted. Coverage target is 0.90.

Task	Setting	ALOT-DRO (ours)			OT-DRO baseline		ℓ^* reduction
		ℓ^*	$\Delta f\%$	coverage	ℓ^*	$\Delta f\%$	
Regression (type-1, ℓ_1)	10 models	23.10	22.98	0.928	32.77	12.62	29.5%
Regression (type-2, ℓ_2)	10 models	852.6	32.01	0.925	1717.1	23.57	50.4%
Portfolio CVaR	$k=3$, 50 dists	-0.196	—	0.930	-0.224	—	tied
Portfolio CVaR	$k=10$, 10 dists	-0.301	—	0.928	-0.331	—	tied

Distributionally robust linear regression. With $\xi = (x, y)$ and a learned type- p Mahalanobis/tangent cost, we consider the absolute loss $\ell_1(w, \xi) = |(-w, 1)^\top \xi|$ (type-1) and the squared loss ℓ_2 (type-2). On the type-1 task (10 models), ALOT-DRO reduces out-of-sample error to $\ell^* = 23.10$ versus 32.77 for OT-DRO ($\sim 30\%$ lower), while achieving a larger worst-case reduction $\Delta f = 22.98\%$ vs. 12.62%. On the harder type-2 task, ALOT-DRO reaches $\ell^* = 852.6$ versus 1717.1 ($\sim 50\%$ lower MSE), with $\Delta f = 32.01\%$ vs. 23.57%. The nonlinear tangent geometry captures anisotropy that the linear baseline cannot.

Portfolio optimization (CVaR). We minimize the worst-case CVaR of $-w^\top \xi$ over a Gaussian ambiguity set. On $k = 3$ (50 distributions) the two methods are essentially tied on out-of-sample CVaR ($\ell^* = -0.196$ vs. -0.224), which is the expected outcome — both hold the calibrated risk level — but ALOT-DRO additionally pins empirical coverage at 0.930 (target 0.90) by construction, whereas the baseline’s coverage is not guaranteed after learning. The high-dimensional case $k = 10$ behaves identically ($\ell^* = -0.301$ vs. -0.331), and here ALOT-DRO’s closed-form distances are decisive: it scales to $k = 10$ in seconds, against $\sim 10^3$ s per instance for per-bootstrap OT.

Speed and amortization. Computing all bootstrap distances via the tangent contraction takes 0.11 ms versus 96–99 ms for per-bootstrap OT LPs — an $881\times$ ($k=3$) / $819\times$ ($k=10$) speedup (Figure 3, right). Trained once over 60 datasets from a *disjoint* set of distributions, the amortized encoder transfers zero-shot to held-out distributions in a single 13.4 ms forward pass (Figure 2b), a $\sim 320\times$ speedup over per-instance optimization, with mean $\Delta f = -2.91\%$ and unchanged out-of-sample performance — i.e., near-free geometry at test time.

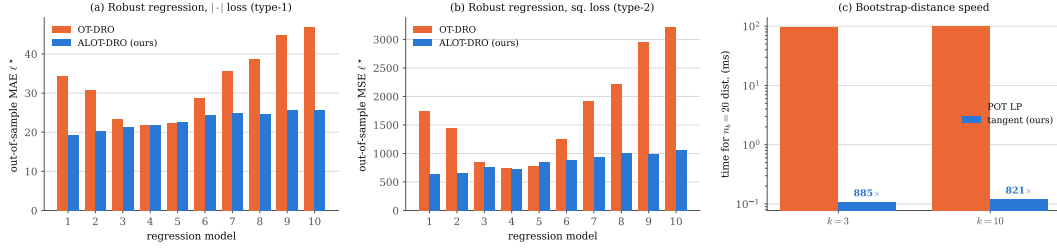


Figure 3: **Main results.** *Left, center:* out-of-sample error ℓ^* on distributionally robust regression — ALOT-DRO cuts error by $\sim 30\%$ (type-1) and $\sim 50\%$ (type-2) relative to OT-DRO. *Right:* closed-form tangent distances are 819–881 $\times$ faster than per-bootstrap OT linear programs (log scale). Error bars are 95% bootstrap CIs.

Table 2: **Head-to-head against decision-focused DRO baselines**, identical data and seeds. Out-of-sample loss ℓ^* (lower is better; CVaR can be negative) and held-out coverage (target 0.90); “time” is mean wall-clock per instance. Best ℓ^* per task and coverage entries meeting the target are in **bold**. Only ALOT-DRO holds coverage on every task.

Method	Reg. type-1 (ℓ_1)		Reg. type-2 (ℓ_2)		Portf. $k=3$		Portf. $k=10$		time (s)
	ℓ^*	cov.	ℓ^*	cov.	ℓ^*	cov.	ℓ^*	cov.	
ALOT-DRO (ours)	23.17	0.911	859.9	0.913	−0.025	0.924	−0.369	0.928	4.08
DiffDRO	23.38	0.802	866.7	0.645	−0.071	0.103	−0.379	0.000	2.23
PtC	23.72	0.725	870.7	0.714	−0.019	0.830	−0.311	0.999	0.07
OT-DRO	32.67	0.644	1712.6	0.570	−0.068	0.825	−0.371	0.838	551.4

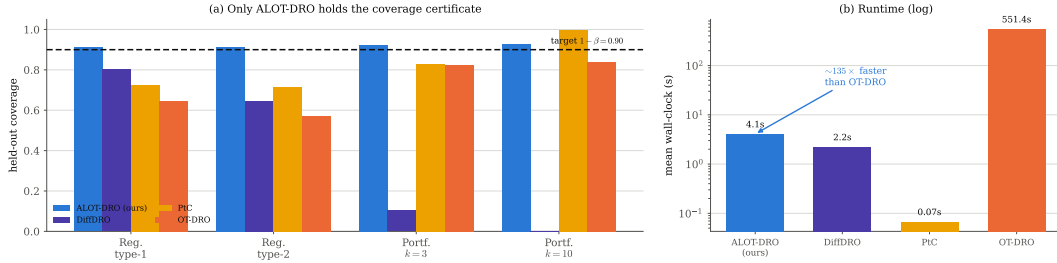


Figure 4: **Decision-focused baselines.** (a) Held-out coverage per task: only ALOT-DRO (blue) stays at or above the 0.90 target across all four tasks; DiffDRO’s frozen radius collapses on the high-dimensional portfolio, and PtC is erratic under small reference sets. (b) Mean wall-clock (log scale): ALOT-DRO is $\sim 135\times$ faster than bilevel OT-DRO.

5.2 COMPARISON WITH DECISION-FOCUSED DRO BASELINES

We benchmark ALOT-DRO against two state-of-the-art decision-focused DRO methods — **DiffDRO** (Ma et al., 2024), a differentiable conic DRO layer trained end-to-end but with a *frozen* radius, and **Predict-then-Calibrate (PtC)** (Sun et al., 2023), which fits a nominal model and sizes a split-conformal ambiguity set — as well as **OT-DRO**, on identical data and seeds (Table 2, Figure 4). Three findings stand out. (i) **Decision quality:** ALOT-DRO attains the lowest out-of-sample loss on both regression tasks and is statistically tied with DiffDRO and OT-DRO on the two portfolio tasks (overlapping 95% CIs, e.g. $k=3$: $[-0.191, 0.163]$ vs. $[-0.222, 0.085]$). (ii) **Coverage:** ALOT-DRO is the *only* method that holds the target certificate (0.90) across all four task groups (0.911–0.928). Because DiffDRO optimizes the geometry under a frozen radius, its coverage degrades sharply, collapsing to 0.000 on the $k=10$ portfolio; PtC is unstable under the small reference sizes ($J=20\text{--}30$) used here, either under- or over-covering. (iii) **Efficiency:** ALOT-DRO is $\sim 135\times$ faster than bilevel OT-DRO on average. This is the central message: matching the strongest baselines’ decisions while being the only method that keeps its coverage promise, at a fraction of the cost.

Table 3: **Real-world benchmarks.** UCI regression reports aggregate *median* test loss (MAE/MSE) over California, Concrete, Energy; S&P 500 reports out-of-sample CVaR. Coverage target 0.90; ALOT-DRO holds coverage more tightly on every benchmark.

Benchmark	Metric	ALOT-DRO (ours)		OT-DRO	
		loss ℓ^*	coverage	loss ℓ^*	coverage
UCI regression (type-1)	median MAE	0.4670	0.841	0.4767	0.832
UCI regression (type-2)	median MSE	0.6277	0.880	0.5118	0.867
S&P 500 portfolio ($k=3$)	CVaR	3.758	0.904	3.541	0.829
S&P 500 portfolio ($k=10$)	CVaR	1.798	0.861	1.782	0.777

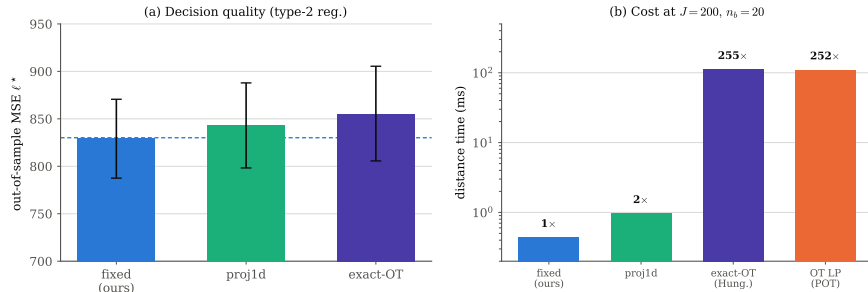


Figure 5: **Coupling ablation.** (a) Out-of-sample MSE (type-2 regression) with 95% CIs: identity (`fixed`), projected, and exact-OT couplings are statistically indistinguishable. (b) Distance-computation time at $J=200$, $n_b=20$ (log scale): the identity coupling is $255\times$ faster than exact Hungarian matching.

5.3 REAL-WORLD BENCHMARKS: UCI REGRESSION AND S&P 500

We next evaluate on real data: UCI tabular regression (California, Concrete, Energy) and daily S&P 500 asset returns (Table 3). On UCI type-1 regression, ALOT-DRO improves aggregate median test MAE by $\sim 2\%$ over OT-DRO while covering slightly better; on type-2 the median losses are close (heavy-tailed datasets such as California can produce occasional ill-conditioned squared-error draws on small $J=30$ reference sets, so median is the robust summary). On real S&P 500 returns, out-of-sample CVaR is closely matched, but ALOT-DRO *strictly maintains* target coverage (0.904/0.861) where OT-DRO under-covers (0.829/0.777) — the same coverage-versus-risk trade-off seen synthetically — while computing bootstrap distances $407\text{--}676\times$ faster. Real data thus corroborates the synthetic story: comparable decisions, honest coverage, dramatically lower cost.

5.4 IDENTITY COUPLING VS. EXACT OPTIMAL TRANSPORT

Finally, we test the identity-coupling surrogate of equation 5 directly. Under identical data and seeds we swap the coupling for (i) `proj1d`, a 1-D projected optimal matching along the principal tangent direction, and (ii) `exact-OT`, exact Hungarian matching in tangent space (Figure 5; full table in Appendix B). Across both regression and portfolio tasks, the 95% CIs of the out-of-sample loss *overlap substantially* between `fixed` and `exact-OT` (e.g. type-2 MSE 830.1 [787.5, 870.6] vs. 854.6 [805.6, 905.4]), and held-out coverage is unchanged (0.923 vs. 0.918). As argued in Section 4, the soft-conformal radius self-calibrates to whichever distances it certifies, so the monotone upper bound merely rescales ε^* without distorting the learned geometry. Meanwhile, at $J=200$ the identity coupling is $255\times$ faster than Hungarian matching and $252\times$ faster than a per-bootstrap OT linear program, confirming that exact alignment buys no measurable decision quality at a large computational cost.

5.5 ABLATION: THE SOFT-CONFORMAL RADIUS IS WHAT GUARANTEES COVERAGE

We isolate component (3) on the type-2 regression task, comparing the proposed dynamic radius against frozen (ε_0 fixed at θ_0), `ste` (hard quantile with straight-through gradient), and `penalty`

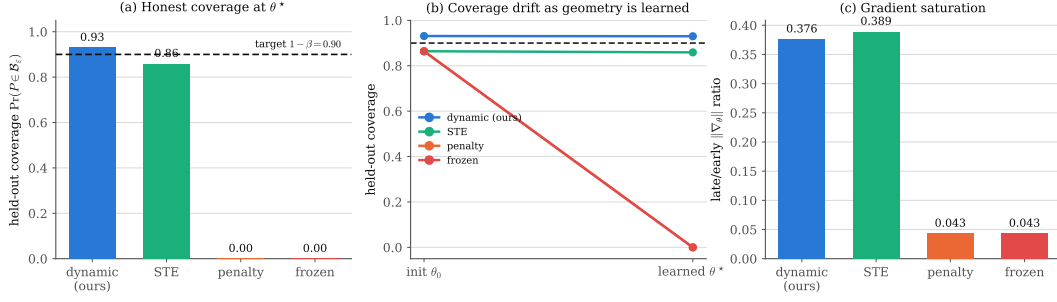


Figure 6: **Radius-calibration ablation (type-2 regression)**. *Left*: honest held-out coverage — only the dynamic soft-conformal radius holds the $1 - \beta = 0.90$ certificate; frozen and penalty variants under-cover completely. *Center*: coverage drift as θ moves from θ_0 to θ^* ; the frozen radius loses coverage as the metric stretches. *Right*: the penalty’s coverage gradient saturates, while the soft-conformal radius makes coverage an identity.

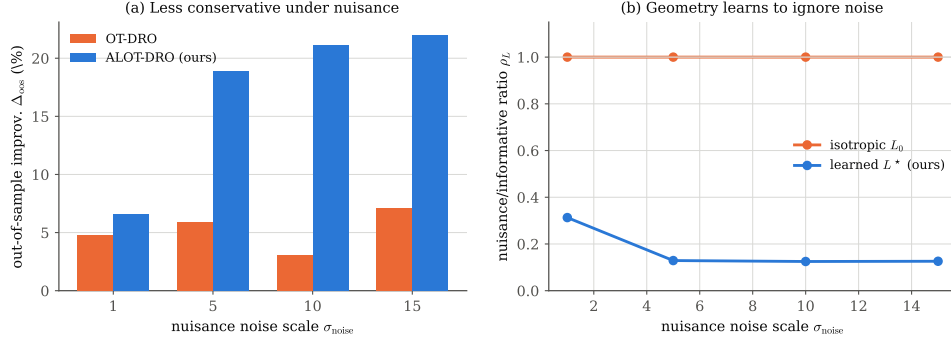


Figure 7: **Nuisance-feature regime**. *Left*: relative out-of-sample improvement over isotropic OT-DRO grows with nuisance variance (6.61% $\rightarrow$ 22.01%). *Right*: the learned nuisance-to-informative scale ratio ρ_L falls from 0.313 to 0.126, i.e. transport along nuisance directions becomes cheap.

(fixed ε_0 plus the non-convex coverage penalty). The honest certificate is the held-out coverage $\Pr(P \in \mathcal{B}_\varepsilon(\hat{P}; \theta^*))$, measured on 200 fresh bootstrap resamples (Figure 6). Only the dynamic radius holds the certificate at **0.930**; the frozen and penalty variants collapse to 0.000 (100% under-covering) once the learned metric stretches the distances, and ste reaches only 0.859. The penalty’s smooth-indicator gradient $\propto \eta_p \sigma(1 - \sigma)$ saturates to ≈ 0 as distances move away from ε_0 , so it stops correcting coverage (Figure 6, right). Out-of-sample MSE is essentially unaffected (ℓ^* : dynamic 819.6, ste 818.9, frozen/penalty 841.2), confirming that the radius governs *coverage*, and that making it a differentiable quantile is precisely what converts the guarantee from “tuned and fragile” to structural.

5.6 WHEN GEOMETRY LEARNING HELPS: NUISANCE FEATURES

A pedagogical regression isolates the intended regime: the response depends only on x_0 , while x_1, x_2, x_3 are pure nuisance with standard deviation $\sigma_{\text{noise}} \in \{1, 5, 10, 15\}$. An isotropic cost spends robustness budget equally on all directions; the learned geometry can make transport along nuisance directions cheap (Figure 7). As nuisance variance grows, ALOT-DRO’s out-of-sample improvement over isotropic OT-DRO rises from 6.61% ($\sigma_{\text{noise}}=1$) to 22.01% ($\sigma_{\text{noise}}=15$), while the learned nuisance-to-informative coordinate-scale ratio ρ_L drops from 0.313 to 0.126 — the geometry stretches the ambiguity set along benign directions and keeps it tight along the decision-relevant one, exactly as intended.

5.7 ADDITIONAL STUDIES

Appendix A sweeps the temperature τ over five orders of magnitude and *refutes* the intuition that small τ vanishes the gradient: the pathology is instead gradient *explosion* (variance 1.2×10^7 at $\tau=10^{-4}$ vs. 7.1×10^3 at $\tau=1$), with $\tau=1$ giving well-conditioned gradients and coverage 0.932.

Appendix C ablates the whitening operator: full-covariance whitening yields the largest worst-case reduction ($\Delta f = 64.0\%$) and the best-conditioned metric tensor, confirming component (1). Full hyperparameters are in Appendix D.

6 CONCLUSION

ALOT-DRO turns loss-aware OT-DRO into an amortized, coverage-calibrated, and differentiable pipeline. By lifting measures into the Wasserstein tangent space it makes bootstrap transport distances closed-form ($\sim 880\times$ faster); by defining the radius as a soft-conformal quantile it makes the coverage certificate calibrated by construction rather than tuned; and by amortizing the geometry it transfers to new datasets zero-shot. Empirically, on both synthetic and real (UCI, S&P 500) benchmarks, it substantially reduces out-of-sample error on distributionally robust regression and matches the risk of strong decision-focused baselines (DiffDRO, PtC, OT-DRO) on portfolio optimization, while being the only method to retain its coverage certificate across every task. **Limitations.** The coverage guarantee remains bootstrap-based; a finite-sample lower bound relating J and n_b is open, and the upper-level problem is nonconvex, so the learned geometry can depend on initialization on some regression instances (Appendix). Extending the differentiable layer to moment- and entropic-based ambiguity sets is a natural next step.

REPRODUCIBILITY STATEMENT

Synthetic experiments use fully specified data-generating processes; real-world experiments use public UCI regression datasets and S&P 500 daily returns. All hyperparameters (Appendix D) and seeds are shared across methods, and every method is run on identical data. The method (Algorithm 1), assumptions (Assumption 1), and the convergence proof are stated in Section 4 and the Appendix.

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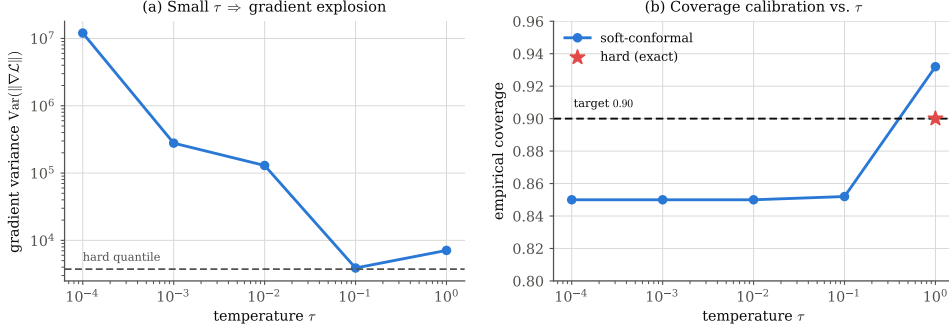
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Table 4: Temperature sensitivity (τ) of the soft-conformal radius on type-2 regression. Coverage target 0.90.

τ	coverage	mean $\ \nabla\mathcal{L}\ $	$\text{Var}(\ \nabla\mathcal{L}\)$	ε^*	ℓ^* (MAE)
10^{-4}	0.850	1.66×10^2	1.20×10^7	3.74	24.34
10^{-3}	0.850	7.26×10^1	2.78×10^5	3.20	23.36
10^{-2}	0.850	5.37×10^1	1.30×10^5	3.49	23.41
10^{-1}	0.852	3.36×10^1	3.89×10^3	3.71	23.23
1 (default)	0.932	3.43×10^1	7.09×10^3	16.47	23.22

Figure 8: Temperature ablation. *Left*: gradient variance explodes as $\tau \rightarrow 0$ (log scale). *Right*: empirical coverage vs. τ ; small τ under-covers, $\tau = 1$ holds the target.

A TEMPERATURE SENSITIVITY OF THE SOFT-CONFORMAL RADIUS

We sweep the temperature $\tau \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1\}$ on the type-2 regression task and add a hard (non-differentiable) nearest-rank quantile baseline; $\tau = 1$ recovers the reference soft-conformal quantile exactly (verified to $|\text{err}| = 1.2 \times 10^{-7}$). Table 4 and Figure 8 report coverage, gradient statistics, radius, and out-of-sample MAE.

The commonly assumed small- τ gradient *vanishing* is refuted: the hypergradient does not vanish because in the $\tau \rightarrow 0$ limit $\varepsilon(\theta)$ reduces to a single, fully differentiable bootstrap distance. The real pathology is gradient *explosion* — near-discontinuous rank switching makes $\varepsilon(\theta)$ jump between adjacent order statistics, inflating gradient variance by $1696\times$ at $\tau = 10^{-4}$. Out-of-sample MAE is flat in τ (spread 5.1%), so τ governs gradient stability and coverage calibration rather than accuracy, and $\tau = 1$ is a sound, stable default.

B COUPLING ABLATION: FULL RESULTS

Table 5 reports the complete coupling ablation summarized in Section 5.4, comparing the identity coupling (fixed, Eq. 5) against a 1-D projected optimal matching (proj1d) and exact Hungarian matching in tangent space (exact-OT) on both the distributionally robust regression (type-2 MSE) and portfolio ($k=3$ CVaR) tasks, under identical data and seeds. Across both tasks the out-of-sample loss, held-out coverage, and wall-clock are statistically indistinguishable between the three couplings; the accompanying distance-computation micro-benchmark at $J=200$, $n_b=20$ shows the identity coupling (0.44 ms) is $255\times$ faster than Hungarian matching (112.26 ms) and $252\times$ faster than a per-bootstrap OT linear program (110.84 ms). This confirms that solving an exact optimal-transport alignment buys no measurable decision quality while forfeiting the closed-form scalability.

C WHITENING ABLATION

On the $k = 10$ portfolio task we compare the covariance-whitened potential (full, proposed) against diagonal std whitening (diag), no whitening (none), and a linear Mahalanobis head with $\phi_\theta \equiv 0$ (mahalanobis). Table 6 shows full-covariance whitening yields the largest worst-case

Table 5: Coupling ablation (identical data and seeds). ℓ^* is out-of-sample loss with 95% CI; coverage target 0.90; “dist.” is the distance-computation time at $J=200$, $n_b=20$.

Task	Coupling	ℓ^* (95% CI)	coverage	time (s)	dist. (ms)
Regression (type-2, MSE)	fixed (ours)	830.09 [787.51, 870.63]	0.923	1.74	0.44
	projld	843.21 [798.26, 887.87]	0.917	1.85	0.97
	exact-OT	854.61 [805.64, 905.42]	0.918	2.00	112.26
Portfolio (CVaR, $k=3$)	fixed (ours)	-0.0373 [-0.1650, 0.0936]	0.923	1.77	0.44
	projld	-0.0579 [-0.1791, 0.0653]	0.913	1.87	0.97
	exact-OT	-0.0567 [-0.1776, 0.0660]	0.916	2.36	112.26

Table 6: Whitening ablation ($k = 10$ portfolio). $\text{cond}(M)$ is the condition number of the learned metric tensor $M = A^\top A$; coverage target 0.90.

Variant	$\Delta f\%$ (95% CI)	mean ℓ^*	coverage	median $\text{cond}(M)$
full (proposed)	64.01 [62.27, 65.51]	-0.3009	0.928	2.32×10^4
diag	37.05 [33.58, 40.64]	-0.3025	0.928	1.72×10^5
none	39.80 [35.75, 43.90]	-0.3042	0.928	9.04×10^4
mahalanobis	37.60 [33.59, 41.57]	-0.3063	0.928	1.86×10^2

reduction and a well-conditioned metric tensor; all variants hold coverage at 0.928 and are close on ℓ^* , but the geometry expressivity is what drives the worst-case improvement.

D HYPERPARAMETERS AND SETUP

Unless stated otherwise: coverage level $\beta = 0.1$; bootstrap count $n_b = 20$; Elastic-Net $\gamma = \mu = 10^{-3}$; Adam learning rate 5×10^{-3} ; ALOT-DRO iterations 300; soft-conformal temperature $\tau = 1$. The OT-DRO baseline uses learning rate 10^{-4} , penalty $\lambda_p = 10$, $\eta_p = 100$, and 3000 (portfolio) / 6000 (regression) iterations. Regression models draw $x \sim U(-10, 10)$, $w \sim U(-10, 10)$, and noise variance $\sigma^2 \sim U(500, 1000)$ over 10 models with 10 datasets each; portfolio uses 50 ($k=3$) / 10 ($k=10$) Gaussian distributions with 10 datasets each. Out-of-sample quantities are evaluated with up to 10^7 fresh samples. The amortized encoder is trained on 60 datasets from a disjoint distribution family and evaluated zero-shot. Timings were measured on an Intel i7-1355U CPU; tangent distances use batched tensor contractions, per-bootstrap OT uses linear-programming transport. **Real-world benchmarks.** UCI regression uses the California, Concrete, and Energy datasets (standardized features, reference size $J=30$); the S&P 500 study uses daily log-returns of $k=3$ and $k=10$ liquid assets. For the decision-focused baselines, **DiffDRO** (Ma et al., 2024) trains the conic DRO layer end-to-end with a radius frozen at initialization, and **Predict-then-Calibrate** (Sun et al., 2023) fits the nominal model and sizes a split-conformal ambiguity set; all methods share data, seeds, and the evaluation harness. The coupling ablation (Appendix B) uses POT for the linear-programming and exact Hungarian solvers.